NovaMart — Real-World Data Analyst Workflow

1. Define the Business Problem

Understand what decision the analysis should support, identify stakeholders, define success metrics, and clarify the required time period and level of detail.

2. Understand the Data

- Identify all files/tables
- Understand each table's business meaning
- Inspect rows, columns and data types
- Identify primary and foreign-key-like fields
- Understand grain: what does one row represent?

3. Load the Data

Load raw files into Python/Pandas or a database. Keep an untouched raw copy so every transformation can be traced back.

4. Profile the Data

- Shape of each table
- Column names and types
- Missing values
- Duplicate rows
- Unique values
- Minimum/maximum
- Mean/median and distributions
- Potential anomalies

5. Validate Data Quality

- Check required fields
- Check uniqueness of identifiers
- Check referential relationships
- Check date ranges
- Check impossible or suspicious values
- Check numerical consistency
- Compare totals with source reports when available

6. Clean the Data

- Fix data types
- Standardize text values
- Handle missing values appropriately
- Remove true duplicates
- Correct formatting issues
- Validate outliers rather than automatically deleting them

7. Transform & Prepare

Create calculated fields and analytical tables needed for the business questions. Examples include revenue, cost, profit, profit margin, customer groups, product groups, and time dimensions.

8. Analyze

Start with descriptive analysis, then segment by meaningful dimensions. Compare totals, trends, distributions, contribution, and efficiency. Look for differences between high-volume and high-margin segments.

9. Visualize

- Choose charts based on the question
- Use line charts for trends
- Use bars for comparisons
- Use KPI cards for headline metrics
- Use stacked charts when composition matters
- Avoid unnecessary visual decoration

10. Build the Dashboard

Organize dashboards by business purpose. A strong structure is Executive Overview → Detailed Performance → Profitability/Drivers.

11. Dashboard QA

- Check filters
- Verify calculations
- Cross-check totals
- Check sorting
- Check labels
- Check tooltips
- Check number formats
- Check clipping and spacing
- Check that every chart has a business purpose

12. Communicate Insights

Summarize what changed, where it happened, why it matters, and what action a stakeholder could consider. Avoid simply describing charts.

13. Deliver

- Final dashboard
- Project summary
- Methodology/workflow notes
- Metric definitions
- Key findings
- Limitations and assumptions